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(54) **CLEAN ROOM**

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Primary Examiner — Jorge A Casanova

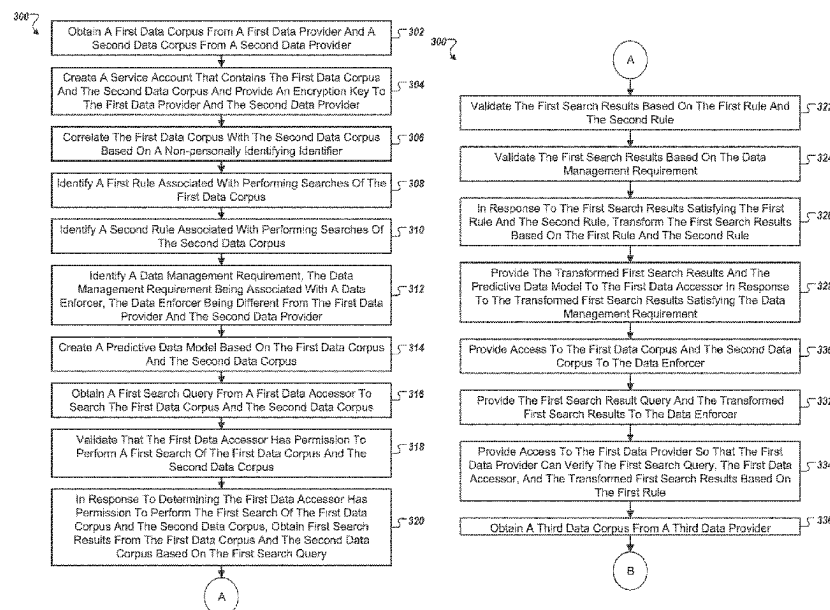
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(57)

ABSTRACT

Embodiments of the present disclosure may provide a data clean room allowing secure data analysis across multiple accounts, without the use of third parties. Each account may be associated with a different company or party. The data clean room may provide security functions to safeguard sensitive information. For example, the data clean room may restrict access to data in other accounts. The data clean room may also restrict which data may be used in the analysis and may restrict the output. The overlap data may be anonymized to prevent sensitive information from being revealed.

25 Claims, 6 Drawing Sheets



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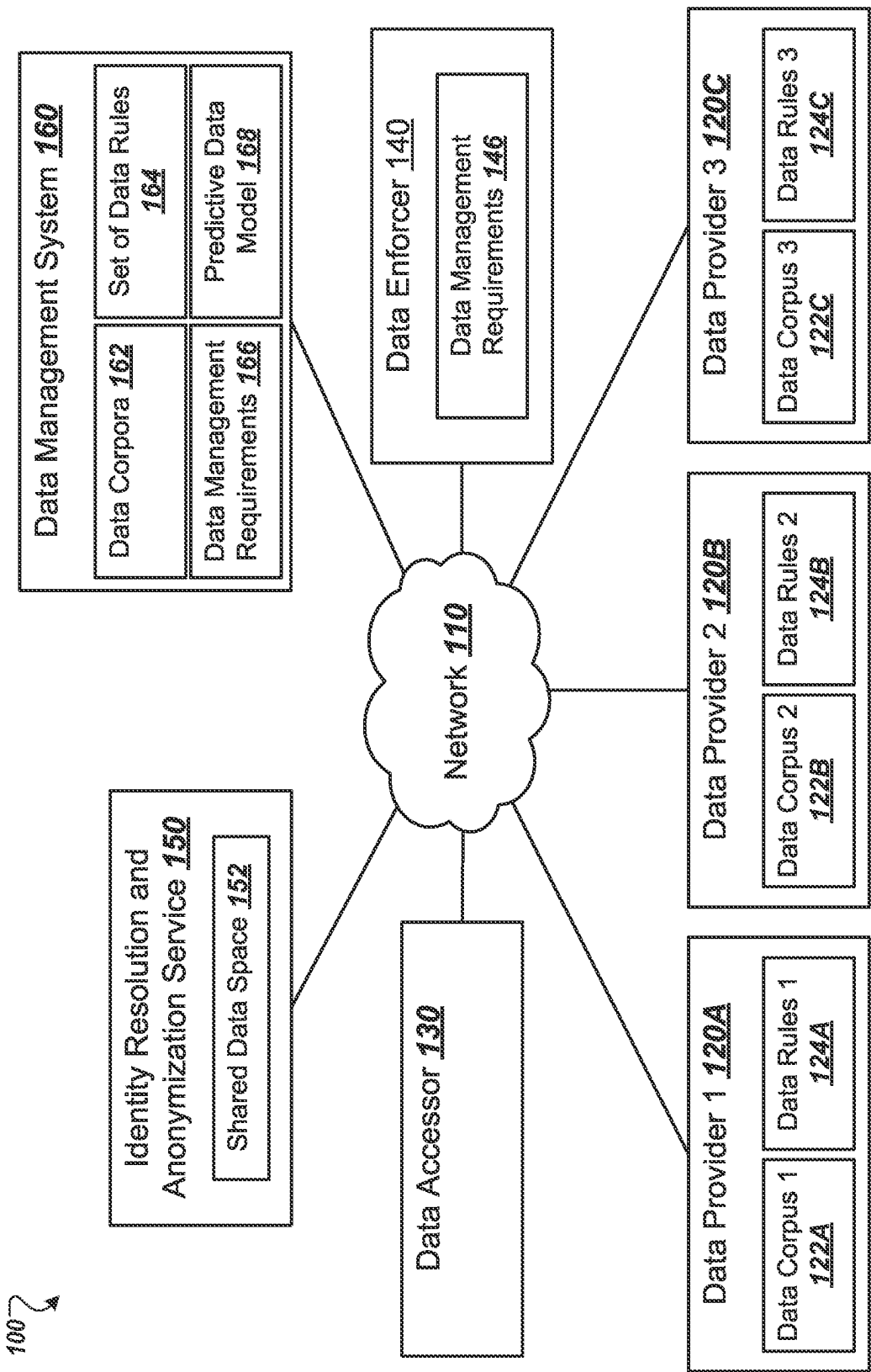


FIG. 1

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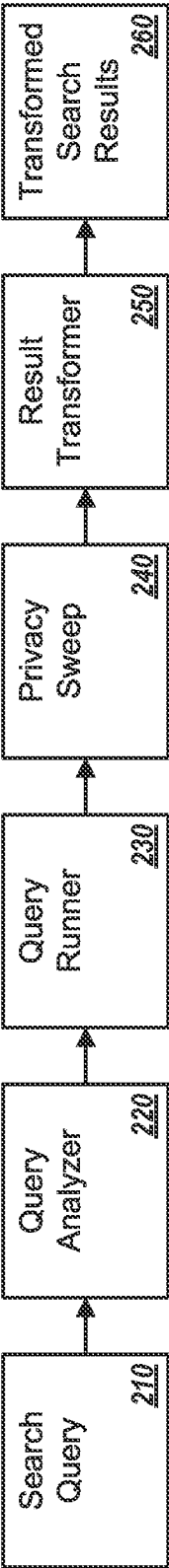


FIG. 2

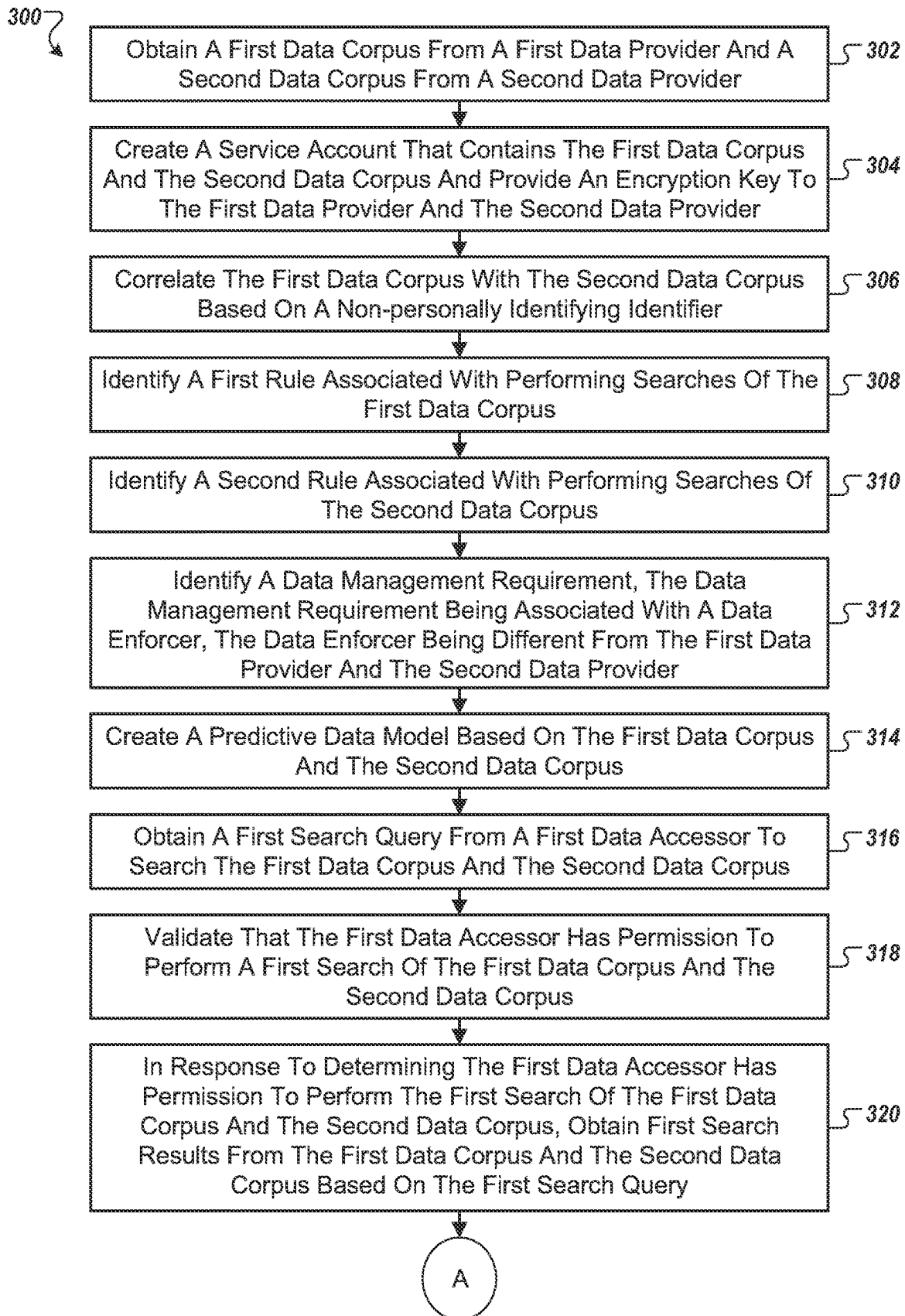


FIG. 3A

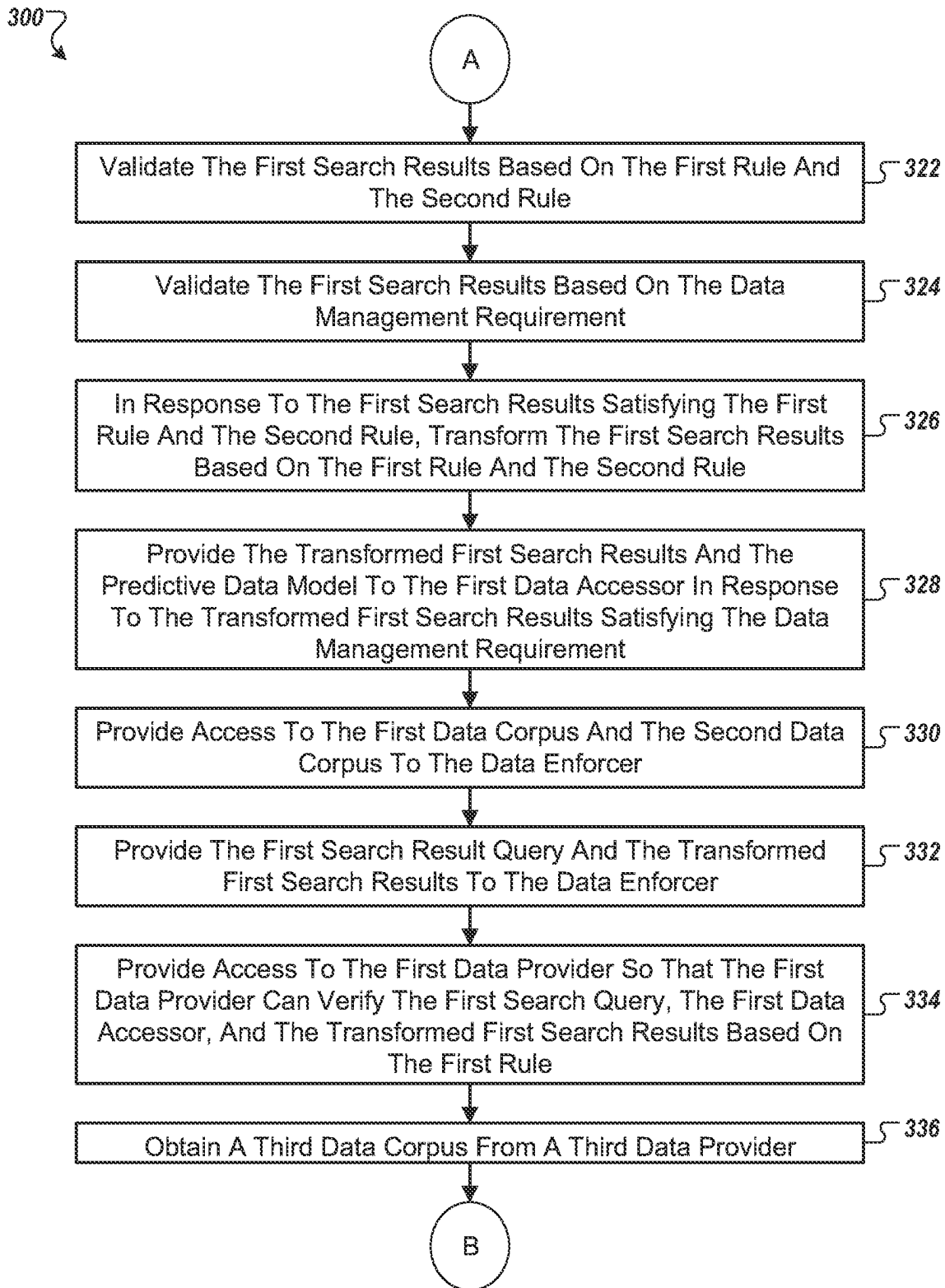
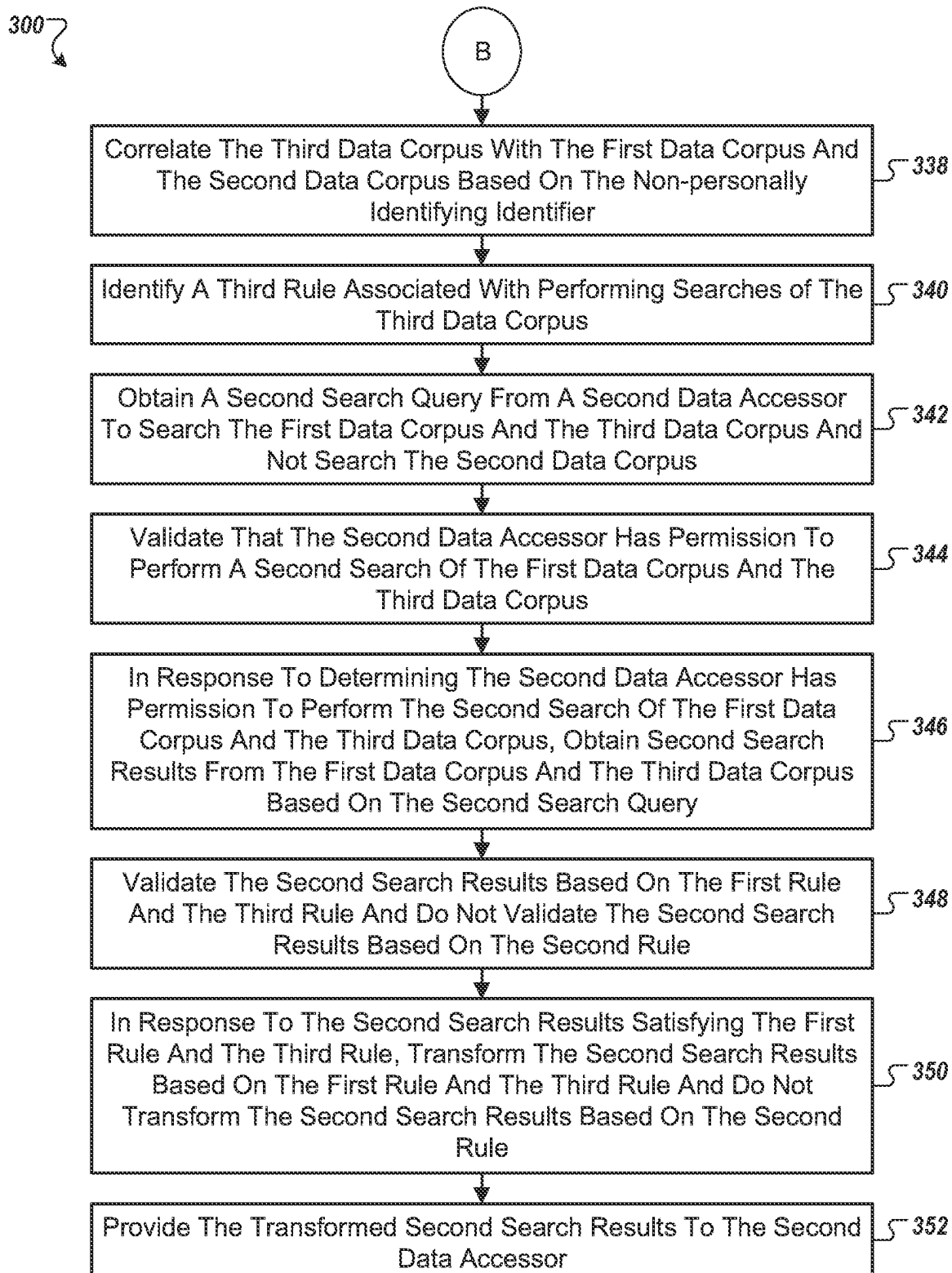


FIG. 3B



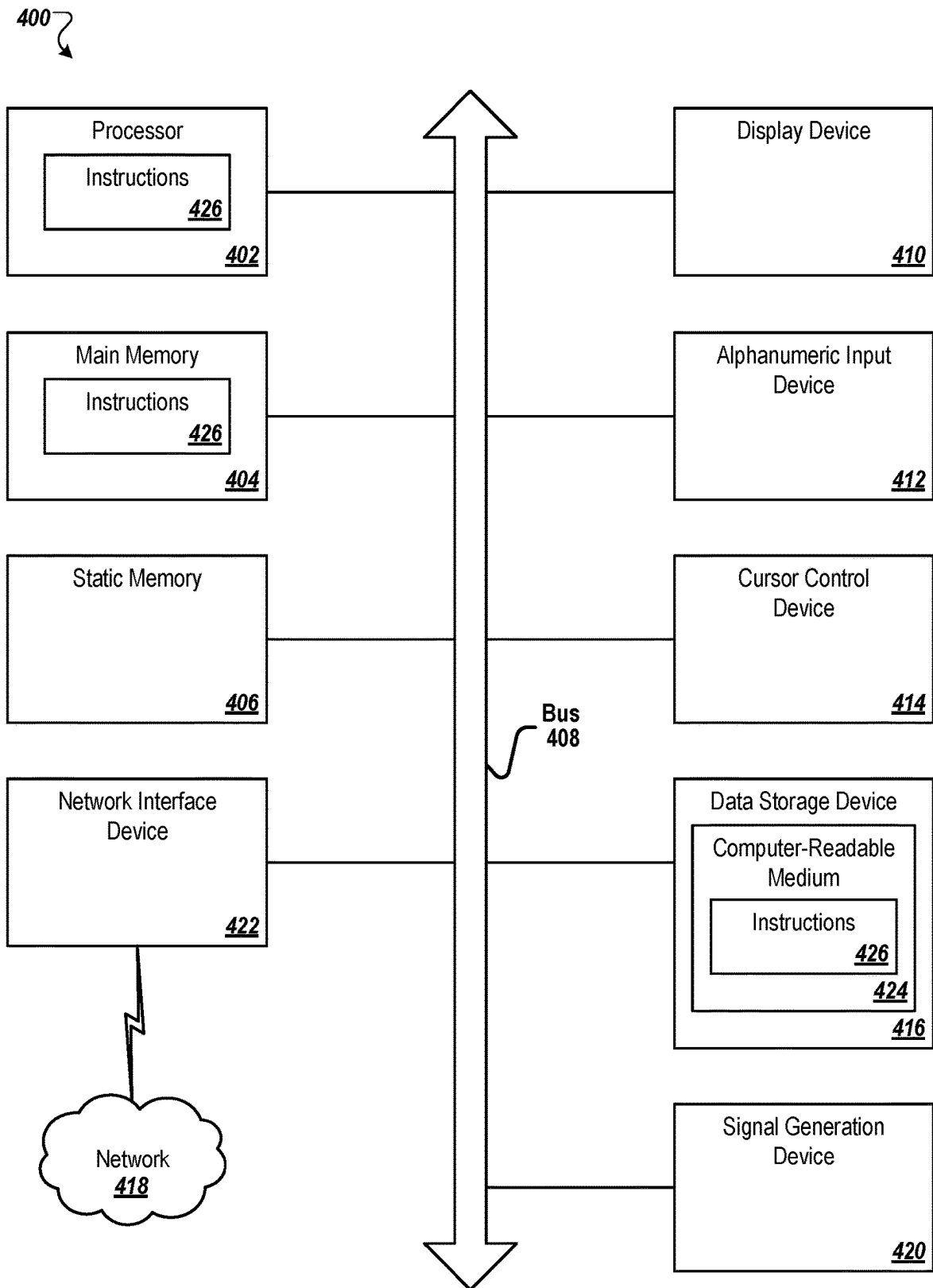


FIG. 4

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CLEAN ROOM

TECHNICAL FIELD

The present disclosure generally relates to securely analyzing data across different accounts using a data clean room.

BACKGROUND

Currently, most digital advertising is performed using third-party cookies. Cookies are small pieces of data generated and sent from a web server and stored on the user's computer by the user's web browser that are used to gather data about customers' habits based on their website browsing history. Because of privacy concerns, the use of cookies is being restricted.

Companies may want to create target groups for advertising or marketing efforts for specific audience segments. To do so, companies may want to compare their customer information with that of other companies to see if their customer lists overlap for the creation of such target groups. Thus, companies may want to perform data analysis, such as an overlap analysis, of their customers or other data. To perform such types of data analyses, companies can use "trusted" third parties, who can access data from each of the companies and perform the data analysis. However, this third-party approach suffers from significant disadvantages. First, companies give up control of their customer data to these third parties, which can lead to unforeseen and harmful consequences because this data can contain sensitive information, such as personal identity information. Second, the analysis is performed by the third parties, not the companies themselves. Thus, the companies have to go back to the third parties to conduct a more detailed analysis or a different analysis. This can increase the expense associated with the analysis as well as add a time delay. Also, providing such information to third parties for this purpose may run afoul of ever-evolving data privacy regulations and common industry policies.

SUMMARY

Embodiments of the present disclosure may provide a data clean room allowing secure data analysis across multiple accounts, without the use of third parties. Each account may be associated with a different company or party. The data clean room may provide security functions to safeguard sensitive information. For example, the data clean room may restrict access to data in other accounts. The data clean room may also restrict which data may be used in the analysis and may restrict the output. The overlap data may be anonymized to prevent sensitive information from being revealed.

BRIEF DESCRIPTION OF THE DRAWINGS

Example embodiments will be described and explained with additional specificity and details through the use of the accompanying drawings in which:

FIG. 1 illustrates an example environment related to an electronic multi-tenant data management system;

FIG. 2 illustrates an example system related to performing a search in an electronic multi-tenant data management system;

FIGS. 3A-3C illustrate example flow diagrams of an example computer-implemented method to perform a search of an electronic multi-tenant data management system; and

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FIG. 4 illustrates an example computing device that may be used with an electronic multi-tenant data management system.

DETAILED DESCRIPTION

The following disclosure sets forth numerous specific details such as examples of specific systems, components, methods, and so forth, in order to provide a good understanding of several embodiments of the present disclosure. It will be apparent to one skilled in the art, however, that at least some embodiments of the present disclosure may be practiced without these specific details. In other instances, well-known components or methods are not described in detail or are presented in simple block diagram format in order to avoid unnecessarily obscuring the present disclosure. Thus, the specific details set forth are merely examples. Particular implementations may vary from these example details and still be contemplated to be within the scope of the present disclosure.

Users generate data across a variety of platforms. Each of these platforms may obtain data relative to particular habits and/or activities of users. For example, web-based shopping sites may obtain a shopping history of a user, a purchase history of a user, a search history of a user, browsing history of a user, and other information. A video streaming service may have a viewing history of a user, a search history of a user, customer ratings submitted by the user, and other information. A social media site may have a list of topics, pages, and/or companies that a user has "liked", subjects and content of posts by a user, a list of topics, pages, and/or companies that a user has "followed", comments submitted by a user, and other information. In today's digital age, users may interact with multiple platforms and services each day. The multiple platforms and services are typically owned and operated by different entities that do not share their data with others. It may be beneficial for companies to be able to search data from multiple different sources to identify a more full picture of user activity, identify trends for a user and among multiple users, improve the targeting of advertising for individuals, and/or measure how successful advertising campaigns are, among others.

However, searching and analyzing data across different companies, platforms, and services may be difficult and/or impossible for a variety of reasons. If user data is not hidden, encrypted or anonymized, companies may be hesitant to share their own data with competitors, particular when the data may help competitors target the companies' customers. For example, a social media site may have little incentive to share its collection of data about users with a video streaming company or a web-based shopping site. Additionally, legal restrictions, including privacy regulations, may regulate the dissemination or use of personally identifying information, preventing one company from sharing information it gathers with other companies.

Aspects of the present disclosure address these and other shortcomings of prior systems by improving the sharing of data across computing systems. The present disclosure provides an electronic multi-tenant data management system that entities can use to cross-share data among other entities, while still maintaining privacy of user information and company proprietary information. Using the electronic multi-tenant data management system, entities can have access to a more full set of data about a user and/or a set of users. This increased access may enable the companies to provide better electronic data services, such as advertising, to users. Additionally, electronic multi-tenant data manage-

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ment systems may facilitate the verification of compliance with regulatory restrictions on the sharing and use of information.

FIG. 1 illustrates an example environment **100** in which embodiments of the present disclosure can be implemented. The environment **100** may include a network **110**, a data provider **1 120A**, a data provider **2 120B**, and a data provider **3 120C** (collectively the data providers **120**), a data accessor **130**, a data enforcer **140**, an identity resolution and anonymization service **150**, and a data management system **160**.

In some embodiments, the network **110** may include a public network (e.g., the Internet), a private network (e.g., a local area network (LAN) or a wide area network (WAN)), a wired network (e.g., an Ethernet network), a wireless network (e.g., an 802.11 network, Bluetooth network, or a Wi-Fi network), a cellular network (e.g., a Long Term Evolution (LTE) or LTE-Advanced network), routers, hubs, switches, server computers, and/or a combination thereof.

Each of the data providers **120**, the data accessor **130**, the data enforcer **140**, the identity resolution and anonymization service **150**, and the data management system **160** may be or include a computing device such as a personal computer (PC), a laptop, a server, a mobile phone, a smart phone, a tablet computer, a netbook computer, an e-reader, a personal digital assistant (PDA), or a cellular phone etc.

Although FIG. 1 depicts three data providers **120**, in some embodiments the environment **100** may include any number of data providers **120**. The data providers **120** may be associated with different entities that may generate and/or obtain data associated with users. For example, the data providers **120** may be associated with video streaming companies, web-based shopping companies, social media companies, search engines, e-commerce companies, and/or other any other type of company. For example, the data provider **1 120A** may be associated with a video streaming company and/or platform, the data provider **2 120B** may be associated with a web-based auction company, and the data provider **3 120C** may be associated with a search engine.

Each of the data providers **120** may be configured to obtain data associated with users of services provided by the data providers **120**. Continuing the above example, the data provider **1 120A** may obtain data associated with a variety of customers as the data corpus **1 122A**. The data corpus **1 122A** may include user names, addresses, billing information, user preferences, user settings, user search histories, user viewing histories, user ratings, etc. For example, the data corpus **1 122A** may include a listing of each video streamed by each user together with a time when each video was streamed, a location where each video was streamed, a number of times each video was streamed, any ratings submitted by a user associated with any videos streamed by the user, searches performed by the user, purchases made by the user, language settings of the user including subtitles, captions, language tracks, and other data of the user. In some embodiments, the data corpus **1 122A** may correlate data with particular users based on a user's name, user identification, email address, billing information, etc.

Similarly, the data provider **2 120B** may obtain data associated with a variety of customers as the data corpus **2 122B**. The data corpus **2 122B** may include similar data as the data corpus **1 122A** but may be associated with, in this example, a web-based auction company. For example, the data corpus **2 122B** may include a listing of each auction that is being tracked by each user, each bid and purchase made by each user, product ratings submitted by each user relative to purchases made by the user, buyer and/or seller ratings associated with each user, searches performed by each user,

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items each user has listed for sale, a user's physical location, etc. In some embodiments, the data corpus **2 122B** may correlate data with particular users based on a user's name, user identification, email address, billing information, etc.

Similarly, the data provider **3 120C** may obtain data associated with a variety of customers as the data corpus **3 122C**. The data corpus **3 122C** may include similar data as the data corpus **1 122A** and the data corpus **2 122B** but may be associated with, in this example, a search engine. For example, the data corpus **3 122C** may include a listing of each search performed by each user, the sequence of each search performed by each user, a timing of each search performed by each user, each search result that is examined by each user (e.g. each search result that is opened, read, clicked, etc.), and other data. In some embodiments, the data corpus **3 122C** may correlate data with particular users based on a user's name, user identification, email address, billing information, etc.

The data corpora **122** may additionally include other information such as, for example, tracked locations of user input (e.g., tracking where a user clicks, where a user moves a mouse, where a user drags a finger on a touchscreen), tracked keystrokes of users, tracked eye movement and eye focus of users, advertisements that are visited by each user, purchase and return history for each user, location of users, demographic information about users such as the users age, ethnicity, education level, income level, gender, etc. and other user data.

In situations in which the systems discussed here collect personal information about users, or may make use of personal information, the users may be provided with an opportunity to control whether programs or features collect user information (e.g., information about a user's social network, social actions, interactions or activities, profession, a user's preferences, a user's viewing history, or a user's current location), or to control whether and/or how to receive content from the content server that may be more relevant to the user. In addition, certain data may be treated in one or more ways before it is stored or used, so that personally identifiable information is removed. For example, a user's identity may be treated so that no personally identifiable information can be determined for the user, or a user's geographic location may be generalized where location information is obtained (such as to a city, ZIP code, or state level), so that a particular location of a user cannot be determined. Thus, the user may have control over how information is collected about the user and used by a content server.

The data corpora **122** may be shared, on a full or limited basis, to the data management system **160**. Each of the data providers **120** may also include corresponding data rules **124** that dictate how the respective data corpus may be shared, used, access, etc. by other data providers **120** that can access the data management system. **160**. For example, the data provider **1 120A** may include data rules **1 124A**, the data provider **2 120B** may include data rules **2 124B**, and the data provider **3 120C** may include data rules **3 124C**. The data rules **124** may include restrictions on access to the data corpora **122**. For example, the data rules **1 124A** may include rules established by the data provider **1 120A** for accessing the data corpus **1 122A**. The data rules **1 124A** may include a list of individuals, corporations, and/or entities who may access the data corpus **1 122A** via the data management system **160**. Additionally or alternatively, in some embodiments, the data rules **1 124A** may include a permission list which may grant different individuals, corporations, and/or entities different levels of access to the data corpus **1 122A**.

For example, a first entity may have full access while a second entity may only have access to a subset of the data corpus 1 122A.

The data rules 1 124A may also include privacy requirements. For example, the privacy requirements may include a requirement for a minimum number of user data to be disclosed in response to a search query such as a minimum bin aggregation rule. For example, the minimum bin aggregation may be 100 users. The user data may be shared on an individual basis, or the user data may be aggregated. If a search results in fewer than 100 results, the search results of the data corpus 1 122A may not be disclosed as the number of search results may not satisfy the minimum bin aggregation rule. Additionally or alternatively, if the search results in fewer than 100 results, the search results of the data corpus 1 122A may not be aggregated and the aggregated data may not be shared. In at least one embodiment, user data that is shared is anonymized and personally identifiable user information is removed and/or hidden from being identified by data providers other than the data provider that is sharing the data. In some embodiments, search results may need to satisfy multiple data rules 124 such as the data rules 1 124A and the data rules 2 124B. In these and other embodiments, the data rules 1 124A may include a first minimum bin aggregation rule and the data rules 2 124B may include a second minimum bin aggregation rule. If the first minimum bin aggregation rule is stricter (i.e., greater) than the second minimum bin aggregation rule, the search results may only need to satisfy the first minimum bin aggregation rule.

The data rules 1 124A may also include data transformation rules. For example, the data transformation rules may include a requirement for grouping of search results into bins. For example, in response to a search query, results from the data corpus 1 122A may be grouped into bins of a particular size and/or the number of search results may be rounded to the nearest bin size. When the bin size is 30, the results may be rounded to the nearest 30. Alternatively or additionally, in some embodiments, data transformations may include fuzzing of data. For example, rather than providing exact values for data included in the data corpus 1 122A, the data management system 160 may provide the values of the data modified by a relatively small random amount, or data that has been aggregated.

Each of the data providers 120 may provide its corresponding data corpus 122 and data rules 124 to the data management system 160 and may be subject to the respective data rules 124.

The data accessor 130 may be associated with any entity, including the same or different entities associated with the data providers 120. In some embodiments, the data accessor 130 may be granted permission to perform searches of one or more the data corpora 122 via the data management system 160. In these and other embodiments, the data accessor 130 may be listed as a party that may access the data corpora 122 subject to the data rules 124. In some embodiments, the data accessor 130 may have access to search some data corpora 122 and may not have access to search other data corpora 122. For example, the data rules 1 124A and data rules 2 124B may list the data accessor 130 as an entity that may perform searches of the data corpus 1 122A and the data corpus 2 122B while the data rules 3 124C do not list the data accessor 130 as a permissioned party. Thus, when attempting to perform a search using the data management system 160, the search results may not include results associated with the data corpus 3 122C.

The data enforcer 140 may be associated with a third-party such as, for example, a government entity. For example, the data enforcer 140 may be associated with a regulatory body that works to ensure that data gathered by the data providers 120 and accessed by the data providers 120 and/or the data accessor 130 conform to data management requirements 146. For example, in some jurisdictions, the data management requirements 146 may not permit the gathering of data from minors without consent. Alternatively, in some embodiments, the data management requirements 146 may not permit targeted advertising to minors or to others. Additionally or alternatively, in some jurisdictions, data management requirements 146 may not permit the dissemination of personally identifying information by the party that gathered it to other parties. For example, in some jurisdictions, the data management requirements 146 may allow the data provider 1 120A to gather personally identifying information for use in billing, providing services, etc. but may not allow the data provider 1 120A to sell or distribute that data to other parties. The data enforcer 140 may use the data management system 160 to verify compliance with the data management requirements 146.

The identity resolution and anonymization service 150 may be configured to obscure and/or remove any personally identifying information of the data corpora 122 prior to transmittal of the data corpora 122 to the data management system 160. In some embodiments, the identity resolution and anonymization service 150 may associate the data of the data corpora 122 with an identifier through a process (e.g., a one-way process) such that information from two different data corpora 122 associated with a particular individual may be correlated with each other without revealing the identity of the particular individual. For example, the identity resolution and anonymization service 150 may anonymize and/or remove from the data corpora 122 names, physical addresses, Internet Protocol (IP) addresses, phone numbers, email addresses, credit records, billing information, etc. In some embodiments, the identity resolution and anonymization service 150 may anonymize the data corpora 122 such that the anonymized identifier of a particular user is the same across each of the data corpora 122 in which the particular user's data appears. In some embodiments, the identity resolution and anonymization service 150 may use a live random access memory (RAM) internal identification to generate the anonymized identifier.

In some embodiments, the identity resolution and anonymization service 150 may attempt to protect personally identifiable information by being configured to act as a service account 152 with restricted access. For example, two data providers 120 may desire to share their respective data corpora 122 with one another. The two data providers 120 may then enter into a contract to share data. Responsive to receiving a request from both data providers 120 to create a shared data space 152, the identity resolution and anonymization service 150 may create the shared data space 152. The shared data space 152 may be accessed using one or more of a service account and an encryption key. The shared data space 152 may include some or all of the respective data corpora 122 from both of the data providers 120. Access to the shared data space 152 may be restricted using the service account. Additionally or alternatively, access to the shared data space 152 may be restricted using the encryption key. The encryption key, for example, may limit access only to those data providers 120 that have entered into a contract with one another. Further, an encryption key may only provide one-way access to the data providers 120 that have access to the key. Additionally, an encryption key may be

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generated by Hash-based Message Authentication Code (HMAC), Advanced Encryption Standard (AES), Rivest-Shamir-Adleman (RSA), Triple Data Encryption Standard (TripleDES), or any other method for encrypting data. Data providers **120** that have an encryption key and access to a service account **152** may desire to have additional data providers **120** and their data corpora **122** joined to the service account **152**. In such scenario, a third data provider **120** may be provided an encryption key that grants access to the service account **152** already created by the first two data providers **120**. In at least one embodiment, the encryption key is shared after permission is given by all data providers that currently have access to the encryption key.

The data management system **160** may be configured to receive the data corpora **122** from each of the data providers **120** and correlate the data corpora **122** with each other as the data corpora **162**. In some embodiments, the data management system **160** may obtain the data corpora **122** after the identity resolution and anonymization service **150** has anonymized any personally identifying information from the data corpora **122**. In some embodiments, the data corpora **162** may include an identification of the source of the data, i.e. whether a particular data corpus of the data corpora **162** came from data provider 1 **120A**, data provider 2 **120B**, and/or data provider 3 **120C**. The data management system **160** may identify and correlate data associated with a user, or a group of users in the data corpora **162** and store the correlated data as a searchable record or index.

In some embodiments, the data management system **160** may correlate the data corpora **122** using a non-personally identifying identifier. For example, each of the data corpora **122** may include multiple groups of data, each group of data associated with a particular non-personally identifying identifier. As described above, the non-personally identifying identifiers may be generated by the identity resolution and anonymization service **150**. The non-personally identifying identifiers may be generated in such a way that the same non-personally identifying identifier is generated for a group of data associated with a particular individual regardless of whether the group of data is in the data corpus 1 **122A**, the data corpus 2 **122B**, or the data corpus 3 **122C**. The data management system **160** may thus correlate the data corpora by identifying a first group of data in the data corpus 1 **122A** associated with a particular non-personally identifying identifier, a second group of data in the data corpus 2 **122B** associated with the same particular non-personally identifying identifier, and a third group of data in the data corpus 3 **122C** associated with the same particular non-personally identifying identifier, and then correlating the first group of data with the second group of data and the third group of data.

The data management system **160** may be configured to obtain the data rules **124** from each of the data providers **120** as the set of data rules **164**. In some embodiments, the set of data rules **164** may include an identification of the source of the data rules, i.e. whether particular data rules of the set of data rules **164** came from data provider 1 **120A**, data provider 2 **120B**, and/or data provider 3 **120C**.

The data management system **160** may be configured to obtain the data management requirements **146** from the data enforcer **140** as the data management requirements **166**.

The data management system **160** may be configured to process, verify, and/or validate search queries received from the data providers **120**, the data accessor **130**, and/or the data enforcer **140** to search the data corpora **162** using the set of data rules **164** and the data management requirements **166** as described below relative to FIG. 2. In some embodiments,

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the data management system **160** may also be configured to grant access to the data enforcer **140** to verify compliance with the data management requirements **166**, to verify the contents of the data corpora **162**.

The data management system **160** may be configured to generate a predictive data model **168** of the data corpora **162**. The predictive data model **168** may be generated using machine learning and predictive analytics on the data corpora **162**. For example, a generative adversarial network (GAN) or a privacy-preserving adversarial network (PPAN) may be applied to the data corpora **162** to generate the predictive data model **168** based on the data corpora **162**. Additionally, the predictive data model **168** may be trained on the real data sets contained in the “virtual clean room” or service account **152**, which may limit access to the predictive data model **168** to those data providers **120** that have an encryption key to the service account **152**, and which may restrict data providers **120** from creating their own model on the actual data in the service account **152**. The predictive data model **168** may be used for data providers **120** to predict behaviors, tendencies, and/or trends related to the data corpora **162** that is aggregated in the data management system **160**. The predictive data model **168** may allow an individual data provider **120** a more accurate predictive model by combining data corpora **162** from more than one different data providers **120**. Additionally, the predictive data model **168** may allow the service account **152** to maintain the privacy of the data corpora **162** by not allowing data providers **120** to develop their own predictive data models on the data corpora **162**. For example, data provider 1 **120A** may provide data corpus 1 **122A** to a service account **152** and data provider 2 **120B** may provide data corpus 2 **122B** to the same service account **152**. A predictive data model **168** may be generated on the combined data corpora **162** that data provider 1 **120A** and data provider 2 **120B** have contributed, without disclosing all the data to either of the data providers **120**. The predictive data model **168** may be more accurate and complete than any one data provider **120** could develop on their own data corpora **122**.

Additions, deletions, and modifications may be made to the environment **100** of FIG. 1. In some embodiments, the environment **100** may include more or fewer than three data providers **120**. Alternatively or additionally, in some embodiments, the environment **100** may not include a data accessor **130** or may include multiple data accessors **130**. Alternatively or additionally, in some embodiments, the data accessor **130** may be the same entity as one or more of the data providers **120**. In some embodiments, the environment **100** may not include the data enforcer **140** or may include multiple data enforcers **140**. For example, in these and other embodiments, the environment **100** may include multiple data enforcers **140** and each data enforcer **140** may correspond with a particular jurisdiction and may include data management requirements **146** associated with the particular jurisdiction.

In some embodiments, the environment **100** may not include the identity resolution and anonymization service **150**. In these and other embodiments, each data provider **120** may perform its own data anonymization to remove personally identifying information from its respective data corpus **122**. Alternatively or additionally, the data management system **160** may perform the removing of personally identifying information from the data corpora **122**.

FIG. 2 illustrates an example system **200** related to performing a search in an electronic multi-tenant data management system. The system **200** may correspond with the data management system **160** of FIG. 1. In some embodi-

ments, the system **200** may include a query analyzer **220**, a query runner **230**, a privacy sweep **240**, and a result transformer **250**.

The query analyzer **220** may include a circuit, code and/or computer instructions configured to operate the query analyzer **220** to receive a search query **210** and analyze the search query **210**. The search query **210** may include a request to search for users of particular services at particular locations. The search query **210** may include a request to search any data corpora such as the data corpora **122** and/or data corpora **162** of FIG. 1. For example, the search query **210** may request a search of the data corpus associated with a particular data provider, such as the data corpus **1122A** associated with the data provider **1120A** of FIG. 1. In some embodiments, the query analyzer **220** may analyze the search query **210** based on a set of data rules, such as the set of data rules **164** of FIG. 1, and based on data management requirements, such as the data management requirements **166** of FIG. 1. For example, a data accessor may submit a search query **210** that may request that a search be performed of the data corpus associated with a particular data provider. However, the data rules associated with the particular data provider may not authorize the data accessor to perform searches of the data corpus. The query analyzer may thus validate whether the data accessor has permission to perform a search of the data corpora listed in the search query **210**.

If the query analyzer **220** determines that the search query **210** is not authorized, the query analyzer **220** may provide a message to the originator of the search query **210** indicating that the search failed and/or was not authorized. If the query analyzer **220** determines that the search query **210** is authorized and/or that the data accessor has permission to perform a search of the data corpora referenced in the search query **210**, the query analyzer **220** may provide the search query **210** to the query runner **230**.

The query runner **230** may include a circuit, code and/or computer instructions configured to operate the query runner **230** to run the search query **210**. The query runner **230** may perform a search using the search query **210** over the associated data corpora. As described, the search query **210** may include a list of data corpora to search and a list of terms, locations, data fields, etc. over which to search. The query runner **230** may perform the search using the search query **210** and may obtain search results from the data corpora referenced in the search query **210** and may provide the search results to the privacy sweep **240**. For example, if the search query **210** includes a particular location and a particular behavior, the query runner **230** may identify all data in the data corpora that include the particular location and the particular behavior.

The privacy sweep **240** may include a circuit, code and/or computer instructions configured to operate the privacy sweep **240**. The privacy sweep **240** may perform one or more operations on the search results to verify conformance with the data rules and/or data management requirements. For example, the number of search results may be lower than a required minimum number of search results as set by a data provider in a data rule. For example, the search query **210** may request to search the data corpus of a first data provider and the first data provider may have a data rule requiring at least one hundred results. If the number of results of performing a search using the search query **210** is less than one hundred, the privacy sweep **240** may indicate that the search using the search query **210** has failed the data rules and may return a message indicating that the search failed to the originator of the search query **210**. In some embodi-

ments, the privacy sweep **240** may perform multiple sweeps of the results of the search query **210**. For example, in some embodiments, the privacy sweep **240** may perform a sweep for each data corpus which was searched using the search query **210**. Thus, the privacy sweep **240** may validate the results of the search query **210** based on rules associated with each data corpus. For example, the search query **210** may request a search of the data corpus from a first data provider, the data corpus from a second data provider, and the data corpus from a third data provider. The privacy sweep **240** may perform a first sweep of the search results using the data rules of the first data provider, a second sweep of the search results using the data rules of the second data provider, and a third sweep of the search results using the data rules of the third data provider. In some embodiments, the privacy sweep **240** may also perform a sweep of the search results using the data management requirements. For example, the privacy sweep **240** may determine whether the search results include any personally identifying information.

If the privacy sweep **240** determines that the search results satisfy all of the relevant data rules and the data management requirements, the privacy sweep **240** may provide the search results to the result transformer **250**. In at least one embodiment, failing to satisfy even one of many rules may cause the privacy sweep **240** to not provide (or not authorize provision of) the search results. The result transformer **250** may perform alterations to the data based on the data rules. For example, the data rules associated with one or more data providers may require that the data be fuzzed. As an example of fuzzing the data, a small and/or random amount may be added or subtracted from at least one portion of the actual search results. For example, if the data from a data provider include ad exposure data and the data provider has a rule that ad exposure data must be fuzzed by five minutes, each search result with ad exposure data may have a random amount from -5 minutes to +5 minutes added to the ad exposure data. Other data may also be fuzzed. For example, ages of individuals may be fuzzed by a year, by two years, or by any number of years. As an additional example of data rules, a data rule may require that data be grouped into buckets of a particular amount. For example, if the search results indicate that 97 users satisfy the search query and the data rule requires buckets of 30, the search results may be transformed to indicate that 90 users satisfy the search query. After performing transformations of the search results, the result transformer **250** may output transformed search results **260**. In some embodiments, the result transformer **250** may provide the transformed search results **260** to the party that provided the search query **210** to the query analyzer **220**.

Alternatively or additionally, in some embodiments, the transformed search results **260** may be used to identify potential targets for advertising. For example, the transformed search results **260** may include demographic information, subjects that are "liked" or "favorited", past purchase information, geographic information, frequency of use, and other information that may be used by a company to devise a marketing strategy. For example, a company may target particular channels, social media sites, and or topics to improve its visibility among segments of the population that are more likely to be interested in its products.

Alternatively or additionally, in some embodiments, the transformed search results **260** may be used in the creation of new products. For example, as described above, the transformed search results **260** may include a viewing history of the movies and/or television shows that individu-

als in a particular demographic have watched. By identifying common movies and/or television shows, a television producer may create a new television series to cater to the particular demographic.

Alternatively or additionally, in some embodiments, the transformed search results **260** may be used to identify segments of the population that may be at risk for physical and/or emotional disorders. For example, the transformed search results **260** may include information about individuals with a particular disorder. Using the customer data associated with these individuals, a health agency may identify particular character traits, interests, purchase histories, streaming histories, or other details that may correlate with the particular disorder.

Alternatively or additionally, in some embodiments, the transformed search results **260** may be provided to a data enforcer, such as the data enforcer **140** of FIG. **1**. The search query **210** may also be provided to the data enforcer. In these and other embodiments, the data enforcer may also have access to the data corpora. The data enforcer may thus verify that the data management requirements are being satisfied.

Alternatively or additionally, in some embodiments, the transformed search results **260** may be provided to a data provider, such as the data provider **1120A** of FIG. **1**. The data provider may have access to verify the search query **210**, the data accessor who requested the search query **210**, and the transformed search results **260**.

In some embodiments, the search query **210** may include a request to search some data corpora but not others. In these and other embodiments, the query analyzer **220** may only validate the search query **210** based on data rules associated with the data corpora that the search query **210** requests to search. Similarly, the privacy sweep **240** may only perform sweeps relative to the data rules associated with the data corpora that are searched. Similarly, result transformer **250** may only transform the search results based on data rules associated with the data corpora that are searched.

FIGS. **3A-3C** illustrate an example flow diagram illustrating a method **300** for performing a search of an electronic multi-tenant data management system. The method may be performed by a circuit and/or processing logic that comprises hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions run on a processor to perform hardware simulation), or a combination thereof. Processing logic can control or interact with one or more devices, applications or user interfaces, or a combination thereof, to perform operations described herein. When presenting, receiving or requesting information from a user, processing logic can cause the one or more devices, applications or user interfaces to present information to the user and to receive information from the user.

For simplicity of explanation, the method of FIGS. **3A-3C** is illustrated and described as a series of operations. However, acts in accordance with this disclosure can occur in various orders and/or concurrently and with other operations not presented and described herein. Further, not all illustrated operations may be required to implement the methods in accordance with the disclosed subject matter. In addition, those skilled in the art will understand and appreciate that the methods could alternatively be represented as a series of interrelated states via a state diagram or events.

At block **302**, processing logic may obtain a first data corpus from a first data provider and a second data corpus from a second data provider. At block **304**, the processing logic may correlate the first data corpus with the second data corpus based on a non-personally identifying identifier. At block **306**, the processing logic may identify a first rule

associated with performing searches of the first data corpus. At block **308**, the processing logic may identify a second rule associated with performing searches of the second data corpus. At block **310**, the processing logic may identify a data management requirement. The data management requirement may be associated with a data enforcer. The data enforcer may be different from the first data provider and the second data provider.

At block **312**, the processing logic may obtain a first search query from a first data accessor to search the first data corpus and the second data corpus. At block **314**, the processing logic may validate that the first data accessor has permission to perform a first search of the first data corpus and the second data corpus. At block **316**, the processing logic may, in response to determining the first data accessor has permission to perform the first search of the first data corpus and the second data corpus, obtain first search results from the first data corpus and the second data corpus based on the first search query. At block **318**, the processing logic may validate the first search results based on the first rule and the second rule. At block **320**, the processing logic may validate the first search results based on the data management requirement. At block **322**, the processing logic may, in response to the first search results satisfying the first rule and the second rule, transform the first search results based on the first rule and the second rule. At block **324**, the processing logic may provide the transformed first search results to the first data accessor in response to the transformed first search results satisfying the data management requirement.

At block **326**, the processing logic may provide access to the first data corpus and the second data corpus to the data enforcer. At block **328**, the processing logic may provide the first search query and the transformed first search results to the data enforcer. At block **330**, the processing logic may provide access to the first data provider so that the first data provider can verify the first search query, the first data accessor, and the transformed first search results based on the first rule.

At block **332**, the processing logic may obtain a third data corpus from a third data provider. At block **334**, the processing logic may correlate the third data corpus with the first data corpus and the second data corpus based on the non-personally identifying identifier. At block **336**, the processing logic may identify a third rule associated with performing searches of the third data corpus. At block **338**, the processing logic may obtain a second search query from a second data accessor to search the first data corpus and the third data corpus and not search the second data corpus. At block **340**, the processing logic may validate that the second data accessor has permission to perform a second search of the first data corpus and the third data corpus.

At block **342**, the processing logic may, in response to determining the second data accessor has permission to perform the second search of the first data corpus and the third data corpus, obtain second search results from the first data corpus and the third data corpus based on the second search query. At block **344**, the processing logic may validate the second search results based on the first rule and the third rule and may not validate the second search results based on the second rule. At block **346**, the processing logic may, in response to the second search results satisfying the first rule and the third rule, transform the second search results based on the first rule and the third rule and may not transform the second search results based on the second rule. At block **348**, the processing logic may provide the transformed second search results to the second data accessor.

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FIG. 4 illustrates a diagrammatic representation of a machine in the example form of a computing device **400** within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, may be executed. The computing device **400** may be a mobile phone, a smart phone, a netbook computer, a rackmount server, a router computer, a server computer, a personal computer, a mainframe computer, a laptop computer, a tablet computer, a desktop computer etc., within which a set of instructions, for causing the machine to perform any one or more of the methodologies discussed herein, may be executed. In alternative embodiments, the machine may be connected (e.g., networked) to other machines in a LAN, an intranet, an extranet, or the Internet. The machine may operate in the capacity of a server machine in client-server network environment. The machine may be a PC, a set-top box (STB), a server, a network router, switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated, the term "machine" shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

The example computing device **400** includes a processing device (e.g., a processor) **402**, a main memory **404** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM)), a static memory **406** (e.g., flash memory, static random access memory (SRAM)) and a data storage device **416**, which communicate with each other via a bus **408**.

Processing device **402** represents one or more processing devices such as a microprocessor, central processing unit, or the like. More particularly, the processing device **402** may be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or a processor implementing other instruction sets or processors implementing a combination of instruction sets. The processing device **402** may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), network processor, or the like. The processing device **402** is configured to execute instructions **426** for performing the operations and steps discussed herein.

The computing device **400** may further include a network interface device **422** which may communicate with a network **418**. The computing device **400** also may include a display device **410** (e.g., a liquid crystal display (LCD) or a cathode ray tube (CRT)), an alphanumeric input device **412** (e.g., a keyboard), a cursor control device **414** (e.g., a mouse) and a signal generation device **420** (e.g., a speaker). In one implementation, the display device **410**, the alphanumeric input device **412**, and the cursor control device **414** may be combined into a single component or device (e.g., an LCD touch screen).

The data storage device **416** may include a computer-readable storage medium **424** on which is stored one or more sets of instructions **426** embodying any one or more of the methodologies or functions described herein. The instructions **426** may also reside, completely or at least partially, within the main memory **404** and/or within the processing device **402** during execution thereof by the computing device **400**, the main memory **404** and the processing device **402** also constituting computer-readable media. The instruc-

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tions may further be transmitted or received over a network **418** via the network interface device **422**.

While the computer-readable storage medium **424** is shown in an example embodiment to be a single medium, the term "computer-readable storage medium" should be taken to include a single medium or multiple media (e.g., a centralized or distributed database and/or associated caches and servers) that store the one or more sets of instructions. The term "computer-readable storage medium" shall also be taken to include any medium that is capable of storing, encoding or carrying a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of the present disclosure. The term "computer-readable storage medium" shall accordingly be taken to include, but not be limited to, solid-state memories, optical media and magnetic media.

In the above description, numerous details are set forth. It will be apparent, however, to one of ordinary skill in the art having the benefit of this disclosure, that embodiments of the disclosure may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form, rather than in detail, in order to avoid obscuring the description.

Some portions of the detailed description are presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the means used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here and generally, conceived to be a self-consistent sequence of steps leading to a desired result. The steps are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the above discussion, it is appreciated that throughout the description, discussions utilizing terms such as "identifying," "obtaining," "correlating," "determining," "validating," "receiving," "generating," "transforming," "requesting," "creating," "uploading," "adding," "presenting," "removing," "preventing," "providing," or the like, refer to the actions and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (e.g., electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

Embodiments of the disclosure also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, or it may comprise a computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a non-transitory computer readable storage medium, such as, but not limited to, any type of disk including floppy disks, optical disks, compact disc read-only memories (CD-ROMs) and magnetic-optical disks, ROMs, RAMs, erasable programmable

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read-only memories (EPROMs), electrically erasable programmable read-only memories (EEPROMs), magnetic or optical cards, flash memory, or any type of media suitable for storing electronic instructions.

The words “example” or “exemplary” are used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs. Rather, use of the words “example” or “exemplary” is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise, or clear from context, “X includes A or B” is intended to mean any of the natural inclusive permutations. That is, if X includes A; X includes B; or X includes both A and B, then “X includes A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form. Moreover, use of the term “an embodiment” or “one embodiment” or “an implementation” or “one implementation” throughout is not intended to mean the same embodiment or implementation unless described as such. Furthermore, the terms “first,” “second,” “third,” “fourth,” etc. as used herein are meant as labels to distinguish among different elements and may not necessarily have an ordinal meaning according to their numerical designation.

The algorithms and displays presented herein are not inherently related to any particular computer or other apparatus. Various systems may be used with programs in accordance with the teachings herein, or it may prove convenient to construct a more specialized apparatus to perform the required method steps. The required structure for a variety of these systems will appear from the description below. In addition, the present disclosure is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the disclosure as described herein.

The above description sets forth numerous specific details such as examples of specific systems, components, methods and so forth, in order to provide a good understanding of several embodiments of the present disclosure. It will be apparent to one skilled in the art, however, that at least some embodiments of the present disclosure may be practiced without these specific details. In other instances, well-known components or methods are not described in detail or are presented in simple block diagram format in order to avoid unnecessarily obscuring the present disclosure. Thus, the specific details set forth above are merely examples. Particular implementations may vary from these example details and still be contemplated to be within the scope of the present disclosure.

It is to be understood that the above description is intended to be illustrative and not restrictive. Many other embodiments will be apparent to those of skill in the art upon reading and understanding the above description. The scope of the disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

What is claimed is:

1. A method, comprising:
providing a first party data in a first account in a network-based data system;

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providing a second party data in a second account in the network-based data system;

executing, by a processor, a secure function using the first party data to generate a first result, including creating links to the first party data and anonymizing identification information in the first party data;

sharing the secure function with the second account, the secure function being executed using the second party data to generate a second result;

generating an anonymized correlation with the first result and the second result;

receiving a query request related to the first party data and the second party data;

based on the anonymized correlation, executing the query request on the first party data; and

identifying a query result of the query request, the query request having been run on the first party data and on the second party data.

2. The method of claim 1, further comprising: restricting the second account from accessing at least one of: an aspect of the secure function, or the first party data.

3. The method of claim 1, further comprising: restricting access to the result when a number of search results is below a minimum bin aggregation rule.

4. The method of claim 1, further comprising: restricting the second account from accessing one or more of: the first party data, the secure function, the first result, the second result, or any query result.

5. The method of claim 1, wherein access restriction is handled using encryption.

6. The method of claim 1, wherein executing the query request includes executing the query request on the first party data and the second party data.

7. The method of claim 1, wherein the query request includes a first query portion to request a search of the first party data, and a second query portion to request a search of the second party data.

8. The method of claim 1, wherein identifying the query result includes combining search results related to the first party data and the second party data.

9. The method of claim 1, wherein identifying the query result includes identifying first search results related to the first party data and second search results related to the second party data.

10. A non-transitory machine-storage medium embodying instructions that, when executed by a machine, cause the machine to perform operations comprising:

provide a first party data in a first account in a network-based data system;

provide a second party data in a second account in the network-based data system;

execute a secure function using the first party data to generate a first result, including creating links to the first party data and anonymizing identification information in the first party data;

share the secure function with the second account, the secure function being executed using the second party data to generate a second result;

generate an anonymized correlation with the first result and the second result;

receive a query request related to the first party data and the second party data;

based on the anonymized correlation and the first party data, execute a first portion of the query request to generate results of the first portion of the query request; and

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identify results of the query request having been run on the first party data and on the second party data.

11. The non-transitory machine-storage medium of claim 10, the operations further comprising to restrict the second account from accessing a code of the secure function.

12. The non-transitory machine-storage medium of claim 10, the operations further comprising to restrict access to the result when a number of search results is below a minimum threshold.

13. The non-transitory machine-storage medium of claim 10, the operations further comprising to restrict access to search results when a number of search results is below a minimum threshold.

14. The non-transitory machine-storage medium of claim 10, wherein the operation to execute the query request includes to execute the query request on the first party data and the second party data.

15. The non-transitory machine-storage medium of claim 10, wherein the query request includes a first query portion to request a search of the first party data, and a second query portion to request a search of the second party data.

16. The non-transitory machine-storage medium of claim 10, wherein the operation to identify the query result includes to combine search results related to the first party data and the second party data.

17. A system comprising: one or more processors of a machine; and a memory storing instructions that, when executed by the one or more processors, cause the machine to perform operations comprising:

provide a first party data in a first account in a network-based data system;

provide a second party data in a second account in the network-based data system;

executing a secure function using the first party data to generate a first result, including creating links to the first party data and anonymizing identification information in the first party data; sharing the secure function with the second account, the secure function being executed using the second party data to generate a second result;

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generate an anonymized correlation with the first result and the second result;

receive a query request related to the first party data and the second party data;

based on the anonymized correlation, execute executing a first portion of the query request on the first party data; and

identify a query result of the query request.

18. The system of claim 17, the operations further comprising to restrict the second account from accessing a code of the secure function.

19. The system of claim 17, the operations further comprising to restrict access to the results when a number of search results is below a minimum bin aggregation rule.

20. The system of claim 17, the operations further comprising to restrict access to search results when number of matches is below a minimum threshold.

21. The non-transitory machine-storage medium of claim 10, wherein the operation to identify the query result includes to identify first search results related to the first party data and second search results related to the second party data.

22. The system of claim 17, wherein the operation to execute the query request includes to execute the query request on the first party data and the second party data.

23. The system of claim 17, wherein the query request includes a first query portion to request a search of the first party data, and a second query portion to request a search of the second party data.

24. The system of claim 17, wherein the operation to identify the query result includes to combine search results related to the first party data and the second party data.

25. The system of claim 17, wherein the operation to identify the query result includes to identify first search results related to the first party data and second search results related to the second party data.

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